Lab Exam-I Computer Networks (CSD304): 3 PM – 4 PM

Total Marks: 10

To answer the questions, attach the screenshot of the captured packets whenever required.

1. Start up the Wireshark packet sniffer, and make sure that "http" is in the display filter, so that only captured HTTP messages will be displayed in the packet-list pane. Enter the following URL into your browser:
   Steps:
   (1) Go to www.mit.edu
   (2) Next go to Education page
   (3) Next go to School of Electrical Engineering & Computer Science

   Answer the following based on the above actions and attach the **screenshot.**

a) How many TCP connections were established?
b) Check whether the query for IP is from cache or dedicated DNS server. If it is from a server, specify the IP address of the server.
c) In the above question specify the number of queries executed to resolve the IP address corresponding the URL.
d) Check whether the requested web-page was obtained from cache or actual server.
e) Count the number of TCP and UDP packets generated to satisfy the request.

(2.5)

2. Open the link: https://www.iiitd.ac.in/

   Analyse the TCP packets and answer the following:

a) What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and www.iiitd.ac.in?
b) What is the relation between the acknowledgement number and sequence number?
c) What is the value of the Acknowledgement field in the SYNACK segment sent by the server?
d) What are the sequence numbers of the first four segments in the TCP connection?
e) When was the ACK for each segment received?
f) Given the difference between when each TCP segment was sent, and when its acknowledgement was received, what is the RTT value for each of the above four segments?
g) What is the EstimatedRTT value after the receipt of each ACK? Assume that the value of the EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation for all subsequent segments.

$$\text{EstimatedRTT} = 0.875 * \text{EstimatedRTT} + 0.125 * \text{SampleRTT}$$

(3.5)

Ans. Initially, EstimatedRTT is the SampleRTT of the first segment means the captured RRT.

If the RTT (means SampleRTT) for 1st segment is 0.02746 seconds then assign it as the initial value of EstimatedRTT, i.e.,

EstimatedRTT = 0.02746 (for 1st segment)

For 2nd segment: We need the value of SampleRTT which is the actual RTT that is captured by subtracting (sendtime – Ack-time). So if it is 0.033 seconds, then next equation to be solved is:

EstimatedRTT = 0.875 * 0.02746 + 0.125 * 0.033

For subsequent segments we can evaluate the EsitmatedRTT in similar way.

3. Run nslookup to obtain the IP address of a website of your choice, also determine the authoritative DNS servers for that website. **(1)**

   Ans: Command to check: **nslookup –type=NS website**

4. Suppose you want to open a webpage and the IP address of the URL is not available in the local server. So the address resolution required through DNS. Suppose x number of DNS servers are visited to resolve the IP address and the corresponding round trip times are R_1, R_2 and R_x. Suppose your webpage contains only one object which could be send in a single packet. Let the RTT between server and client with the object file is Rc which has no transmission delay. Now give the expression for the total time spend to execute the request means time after providing the URL on the browser and obtaining the object. **(1.5)**

   Ans. Time to resolve the IP address is: $R_1 + R_2 + ...+ R_x$

   To get the requested page the total time:

   Rc (to establish the connection)+ Rc (To send request for the page and getting the response).

   The desired expression for total time:

   $$\mathbf{2\ Rc + R_1 + R_2 + ...+ R_x}$$

5. Suppose to access a web page containing 10 object files apart from the base HTML file, we initiated the request from our web browser. Address resolution took the total round trip time as R_{Add} then estimate the total time when
   a) HTTP with persistent TCP connection
   a) HTTP with non-persistent TCP connection
   b) HTTP with non-persistent TCP connection and the browser supporting 9 parallel connections.

   To answer the above questions, assume the round trip time from client to server is Rc. **(1.5)**

Ans. 10 object file + 1 base file = Total 11 requests

a) $\mathbf{R_{Add}}$ + 2 **Rc**(To establish the connection + getting the base HTML page)+ **Rc**(To get remaining 10 object together)

   = $\mathbf{R_{Add}}$ + 3 **Rc**

b) $\mathbf{R_{Add}}$ +11* 2 **Rc**

c) R_{Add} + 2 Rc (to establish connection + getting base HTML file)+2 (1 Rc for first 9 requests in parallel + 1 Rc for remaining 2 requests)* 2Rc (establish connection + getting request) = $\mathbf{R_{Add}}$ + 2 **Rc + 2 *** 2 **Rc**